

VISUAL GEOMETRY MEETS DIFFUSION FOR SPARSE-VIEW NOVEL VIEW SYNTHESIS

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Figure 1: **VGGT-Diff** grounds a pretrained video diffusion prior with geometry-routed visual features for sparse-view novel-view synthesis. It maintains strong target-view fidelity across varying pose difficulties while better preserving source-observed details than diffusion-based baselines.

ABSTRACT

We present **VGGT-Diff**, a geometry-routed multi-view diffusion model for sparse-view novel view synthesis. Existing novel view synthesis (NVS) methods face a fundamental trade-off: reconstruction-based approaches preserve observed geometry but struggle to synthesize unseen regions, while diffusion-based methods provide strong generative priors yet rely on implicit source-to-query correspondence. **VGGT-Diff** bridges these regimes by routing visual geometry latents from VGGT- Ω into a pretrained video diffusion model. Each visual token is associated with a 3D point and confidence, then transformed into query-aligned latent conditions through a confidence-aware Visual Geometry Router (VGR) that preserves front and back surface evidence. These conditions guide joint target-view denoising, while Point-Track Residual Consistency (PTRC) regularizes predicted-clean residuals along reliable 3D tracks, improving multi-view stability. We further introduce uncertainty-aware geometry conditioning, combining training-time regularization with inference-time guidance for improved robustness. Experiments show competitive or state-of-the-art performance across interpolation and extrapolation under different viewpoint difficulties. Our code will be released publicly.

1 INTRODUCTION

Novel view synthesis (NVS) aims to render previously unseen viewpoints of a scene from sparse observations. Existing generalizable methods broadly follow two complementary paradigms. Reconstruction-oriented approaches infer explicit or implicit scene representations, including neural radiance fields, Gaussian primitives, and feed-forward 3D representations (Mildenhall et al., 2021; Kerbl et al., 2023; Charatan et al., 2024; Chen et al., 2024a), while Transformer-based models such as LVSM (Jin et al., 2024), RayZer (Jiang et al., 2025a), LagerNVS (Szymanowicz et al., 2026), and SVSM (Kim et al., 2026) learn view synthesis with reduced explicit 3D inductive bias. These approaches provide strong geometric fidelity when target views are well supported by observations, but sparse inputs inevitably leave parts of the scene unobserved, making deterministic reconstruction

increasingly under-constrained under large viewpoint changes. In unseen regions, predictions can degenerate into patch-like artifacts biased toward colors observed in nearby source views.

A complementary line formulates NVS as conditional generation and exploits image or video diffusion priors to complete unseen content (Watson et al., 2022; Kong et al., 2024; Zheng & Vedaldi, 2024; Müller et al., 2024; Yu et al., 2024; Zhou et al., 2025; Wu et al., 2026a). Such models provide strong generative priors, but often lack explicit geometric conditioning to anchor generation to observed scene structure. As a result, generative priors may override geometry-supported evidence and hallucinate plausible yet inconsistent content, leading to structural drift, unstable occlusions, and cross-view inconsistency under wide-baseline interpolation and extrapolation. Meanwhile, visual geometry foundation models such as VGGT (Wang et al., 2025) and VGGT- Ω (Wang et al., 2026) learn rich multi-view representations encoding appearance, 3D structure, correspondence, and confidence, offering a natural way to ground generative NVS with explicit geometric evidence.

These limitations suggest that generative NVS should combine strong completion priors with explicit yet uncertainty-aware geometric guidance. We introduce *VGGT-Diff*, a geometry-routed multi-view diffusion framework that injects visual geometry latents from VGGT- Ω into a pretrained video diffusion model. Rather than rediscovering geometry from RGB and camera rays alone, *VGGT-Diff* associates visual tokens with 3D locations and confidence to construct spatially aligned conditions for source and query views. Since hard projection is brittle near occlusions and uncertain geometry, we propose a confidence-aware Visual Geometry Router (VGR) that preserves front and back surface evidence together with visibility uncertainty, providing query-aligned guidance without treating reconstructed geometry as a complete scene explanation.

A lightweight input projection maps the concatenated source RGB latents, Plücker ray maps, noisy query latents, and routed geometry conditions into the pretrained DiT latent space, allowing *VGGT-Diff* to reuse the video diffusion prior with minimal architectural modification. *VGGT-Diff* then jointly denoises multiple query views within the same DiT sequence, enabling direct information exchange across targets. Joint generation alone, however, does not guarantee consistency with a shared 3D scene. We therefore introduce Point-Track Residual Consistency (PTRC), which uses VGGT-derived 3D correspondences to align denoising residual errors of the same physical point across generated views, rather than forcing view-dependent features or VAE latents to match. Since geometry recovered from sparse observations is inevitably imperfect, we randomly retain, attenuate, or remove the routed geometry condition during training to prevent over-reliance on uncertain projections. This regularization also enables optional Geometry-Prior CFG at inference, which strengthens geometry-aware denoising and further improves novel-view synthesis quality. In this way, visual geometry both conditions the diffusion process and regularizes multi-view generation.

Together, *VGGT-Diff* combines geometry foundation priors with video diffusion for faithful and generative sparse-view NVS. Our contributions are threefold: **(1) Geometry-routed generation.** We introduce a multi-view diffusion framework whose confidence-aware Visual Geometry Router (VGR) transforms VGGT- Ω features and 3D locations into query-aligned conditions, grounding generative completion in observed scene structure; **(2) Geometry-grounded consistency.** We propose Point-Track Residual Consistency (PTRC), which aligns predicted-clean residuals along reliable 3D correspondences to reduce cross-view drift without suppressing valid view-dependent appearance; and **(3) Uncertainty-aware conditioning.** Geometry Condition Dropout prevents over-reliance on uncertain projections, while Geometry-Prior CFG selectively strengthens query guidance at inference. Extensive experiments demonstrate competitive or state-of-the-art performance across pose difficulties and improved geometric reconstructability of jointly generated views.

2 RELATED WORK

Generalizable and Feed-Forward View Synthesis. Novel view synthesis has progressed from scene-specific representations such as NeRF and Gaussian Splatting (Mildenhall et al., 2021; Barron et al., 2021; Kerbl et al., 2023; Chen et al., 2025; 2026) toward generalizable models that amortize reconstruction across scenes. Early approaches infer radiance fields or image-based representations through pixel-aligned features, multi-view stereo, epipolar reasoning, or learned ray aggregation (Yu et al., 2021; Chen et al., 2021; Wang et al., 2021; 2022). More recent methods reduce explicit 3D inductive bias and learn view synthesis with Transformers, including SRT (Sajjadi et al., 2022), ViewFormer (Kulhánek et al., 2022), LVSM (Jin et al., 2024), Efficient-LVSM (Jia et al., 2026),

and RayZer (Jiang et al., 2025a). In parallel, feed-forward models directly predict renderable 3D representations (Charatan et al., 2024; Chen et al., 2024a; Szymanowicz et al., 2024; Hong et al., 2024; Zhang et al., 2024), while recent works further improve geometry-aware scaling and efficiency through LagerNVS (Szymanowicz et al., 2026), SVSM (Kim et al., 2026), projective conditioning (Wu et al., 2026b), and SHARP (Mescheder et al., 2026). These approaches provide strong geometric fidelity and rendering, but deterministic reconstruction remains under-constrained when sparse observations do not cover content revealed by distant query views.

Generative and Geometry-Guided Novel View Synthesis. Generative NVS uses image or video priors to complete unobserved regions. Early diffusion methods perform pose-conditioned synthesis or model joint multi-view distributions (Watson et al., 2022; Liu et al., 2023; Shi et al., 2023; 2024; Liu et al., 2024; Ye et al., 2024; Kong et al., 2024; Zheng & Vedaldi, 2024), while later methods exploit video priors for stronger cross-view coherence (Kwak et al., 2024; Voleti et al., 2024; Müller et al., 2024; Yu et al., 2024; Zhou et al., 2025). Stronger pixel-space backbones improve end-to-end NVS (Elata et al., 2025), and FrameCrafter (Wu et al., 2026a) adapts pretrained video diffusion to complete unordered posed views. Related work studies hybrid deterministic and generative modeling (Le et al., 2026), test-time video completion (Xu et al., 2026), dynamic arbitrary-view generation (Van Hoorick et al., 2026), and correspondence-supervised diffusion (Kwon et al., 2026). However, source-to-query correspondence often remains implicit, limiting robustness to large view-point changes and cross-view consistency. Geometry foundation models such as DUST3R (Wang et al., 2024b), MAST3R (Leroy et al., 2024), VGGT (Wang et al., 2025), and VGGT- Ω (Wang et al., 2026) encode multi-view geometry, correspondence, and cameras. Their geometric evidence has been combined with Gaussian reconstruction and latent video diffusion (Chen et al., 2024b), sparse-view 3D optimization (Wang et al., 2024a), joint image and geometry diffusion (Kwak et al., 2026), latent generative refinement (Hirschorn et al., 2026), geometry-conditioned video diffusion (Kang et al., 2026), and wide-baseline guidance (Zhou et al., 2026). Unlike methods that construct complete Gaussian or radiance-field scenes, *VGGT-Diff* routes visual geometry features, 3D points, and confidence into query-aligned diffusion conditions and reuses their correspondences to regularize jointly generated views. Geometry therefore guides and constrains the generative prior as uncertainty-aware evidence rather than replacing it with deterministic reconstruction.

3 METHOD

Given sparse observations, novel-view synthesis must preserve the scene evidence visible in the source images while completing unobserved regions. Reconstruction models encode correspondence explicitly but have limited support for such completion; video diffusion models provide strong generative priors but leave source-to-query correspondence largely implicit. *VGGT-Diff* uses geometry to connect these two capabilities. Rather than treating the recovered geometry as a complete scene representation, we use it to route visual evidence into a pretrained multi-view diffusion model and to regularize the generated views along shared 3D point tracks. The pipeline is shown in Fig. 2.

3.1 CAMERA-CONDITIONED MULTI-VIEW DIFFUSION

We consider sparse-view novel-view synthesis from M posed source images to N prescribed query cameras. The observed views and requested cameras are represented as

$$\mathcal{S} = \{(I_i^s, K_i^s, T_i^s)\}_{i=1}^M, \quad \mathcal{Q} = \{(K_j^q, T_j^q)\}_{j=1}^N. \quad (1)$$

Our goal is to synthesize query views $\{\hat{I}_j^q\}_{j=1}^N$. We place source and query views in one diffusion sequence, enabling information exchange across slots while preserving view-specific camera controls. A frozen VAE encodes training views into a clean latent volume $Z_0 = [Z_0^s; Z_0^q]$. Following Wan’s linear flow formulation (Wan et al., 2025), we sample a flow time σ and Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$:

$$X_\sigma = (1 - \sigma)Z_0 + \sigma\epsilon, \quad U = \epsilon - Z_0, \quad \mathcal{L}_{\text{FM}} = \frac{1}{N} \sum_{j=1}^N \|V_{\theta,j} - U_j\|_2^2, \quad (2)$$

where X_σ is the diffusion state and $V_{\theta,j}$ is the predicted velocity for query view j . The loss applies only to query slots. Query images define training targets but are never exposed as conditions.

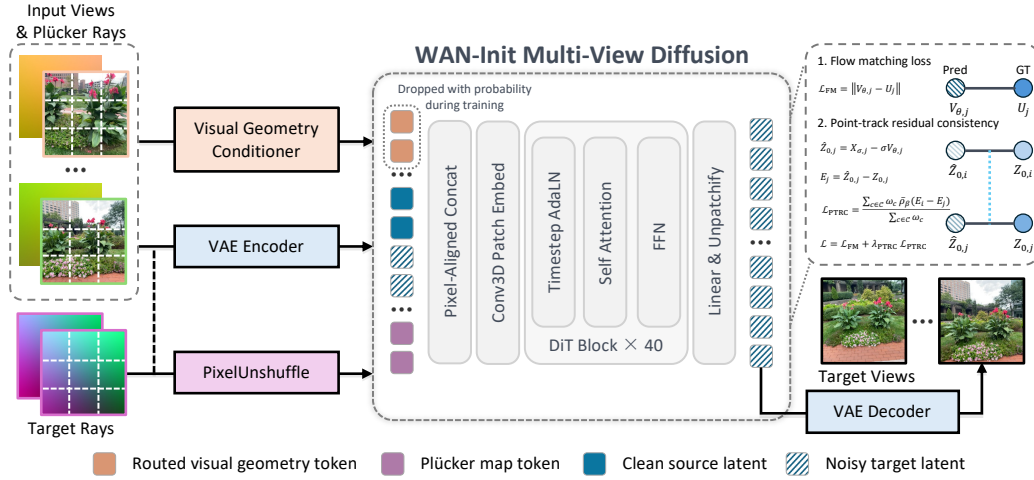


Figure 2: **Overview of VGGT-Diff.** Rather than appending 3D predictions as ordinary input channels, the Visual Geometry Conditioner uses VGGT- Ω points and confidence to route appearance-bearing source features to their corresponding query locations. The diffusion state, clean-source image condition, Plücker rays, and routed visual condition are concatenated before patch embedding and jointly processed by a Wan-initialized multi-view DiT. During training, geometry conditioning is stochastically regularized to prevent over-reliance on uncertain geometry. Flow matching supervises each target prediction, while PTRC aligns predicted-clean residuals along reliable 3D point tracks to improve cross-view consistency.

The model receives three complementary conditions in addition to X_σ . First, Wan’s image-conditioning stream contains clean source latents and blank query slots, providing observed appearance without leaking query RGB. Second, dense Plücker ray maps encode the camera associated with every source and query pixel. Third, the geometry-routed features introduced in Sec. 3.2 provide scene-specific source-to-query correspondence. These signals are concatenated and mapped into the pretrained Wan token space by an expanded input projection. The pretrained image pathway is preserved, while newly introduced camera and geometry inputs are initialized to have no effect at the start of fine-tuning. The source and query tokens are then jointly processed by the Wan DiT. We keep spatial positional encoding but remove temporal ordering from the view axis, since the inputs form a set of camera observations rather than a video timeline. No scene-specific text is used: training and inference share the same fixed empty textual context. After joint denoising, the query latents are decoded independently into the requested views.

3.2 GEOMETRY-ROUTED VISUAL CONDITIONING

Camera rays specify where a view is sampled, but not which source observation supports each query location. As illustrated in Figure 3, we recover this correspondence using a frozen VGGT- Ω (Wang et al., 2026). From source images, VGGT- Ω provides visual features with associated 3D scene points and confidence. A lightweight adapter maps the visual features into the diffusion conditioning space. Rather than appending point coordinates or confidence as ordinary input channels, we use them to determine where each source feature is delivered and how strongly it contributes.

Source and query views follow different alignment paths. Source features already possess reliable native image correspondence and are therefore resampled directly onto the diffusion grid. Reprojecting them through estimated geometry would replace a known association with a noisier one. Query views lack such native correspondence. For a source point X and query camera j , we compute

$$(x^c, y^c, z^c)^\top = R_j X + t_j, \quad \pi_j(X) = \left(f_x \frac{x^c}{z^c} + c_x, f_y \frac{y^c}{z^c} + c_y \right), \quad (3)$$

where R_j and t_j transform the point into query-camera coordinates, and π_j denotes perspective projection under the query intrinsics. The associated visual feature is routed to the resulting query location. This transfers source appearance evidence rather than RGB colors or geometry alone.

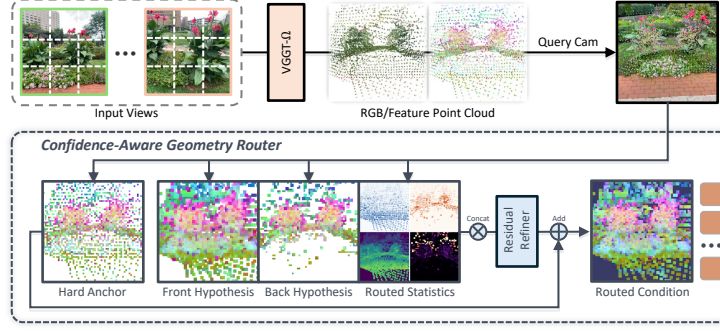


Figure 3: **Visual Geometry Conditioning: Confidence-Aware Routing.** Given source features, 3D points, and confidence from frozen VGGT-Ω, our conditioner projects the points into each query view and routes their features. The residual refiner predicts a correction from front/back hypotheses and routing statistics, then adds it to the hard anchor to form the query-aligned condition.

A hard z-buffer provides a sharp base route, but its surface assignment is unstable near occlusion boundaries: small errors in depth, pose, or sub-pixel projection can switch the selected feature to another surface. A natural remedy is to soften this discrete visibility selection and aggregate all projected features. Although this suppresses abrupt switches, it also blends foreground and background evidence. To preserve sharp surfaces without brittle one-layer assignments, we introduce a confidence-aware geometry router. Let f_n be a projected source feature with continuous latent-grid coordinate (u_n, v_n) , and let $\tilde{c}_n$ denote its normalized VGGT-Ω confidence. For a neighboring grid cell $q = (q_x, q_y)$, let $\mathcal{N}_q$ collect the valid projected points whose bilinear support covers q . We define the splatting weight and aggregated feature of layer ℓ as

$$w_{nq}^\ell = \tilde{c}_n k(u_n - q_x) k(v_n - q_y) \nu_{nq}^\ell, \quad G_q^\ell = \frac{\sum_{n \in \mathcal{A}_q^\ell} w_{nq}^\ell f_n}{\sum_{n \in \mathcal{A}_q^\ell} w_{nq}^\ell + \varepsilon}, \quad k(t) = [1 - |t|]_+. \quad (4)$$

Here, $[a]_+ = \max(a, 0)$, so k is a one-dimensional tent kernel that decreases linearly from one at the projected coordinate to zero one cell away. The separable product of the two kernels distributes a feature over its four neighboring integer cells; before boundary clipping, these spatial weights sum to one. The confidence $\tilde{c}_n$ controls the relative influence of points that meet at the same cell. For the front layer, $\mathcal{A}_q^{\text{front}} = \mathcal{N}_q$, and z_n denotes query-camera depth. We anchor this layer at the nearest splatted depth $z_q^{\text{front}} = \min_{n \in \mathcal{N}_q} z_n$ and define its visibility weight as $\nu_{nq}^{\text{front}} = \exp(-[z_n/z_q^{\text{front}} - 1]_+/\tau)$. It equals one at the front surface and exponentially attenuates deeper points, with τ controlling the decay. For the secondary layer, $\mathcal{A}_q^{\text{back}}$ retains points satisfying $z_n > (1 + \delta)z_q^{\text{front}}$; its nearest point defines the second depth anchor, to which the same soft attenuation is applied. This produces separate front and secondary features without indiscriminately mixing the two surfaces. A lightweight residual refiner receives the front and back hypotheses together with routed statistics describing their support, confidence, and relative depth separation. Its output corrects a depth-selected hard anchor:

$$G_j = G_j^{\text{hard}} + \mathcal{R}(G_j^{\text{front}}, G_j^{\text{back}}, S_j), \quad (5)$$

where S_j collects the routed statistics. The refiner output is initialized at zero, so conditioning begins from the hard geometric route and learns only the corrections supported by layered evidence. Locations without projected support receive no source-specific routed evidence and are completed primarily by the diffusion prior.

3.3 POINT-TRACK RESIDUAL CONSISTENCY

The flow-matching objective in Eq. (2) decomposes over query views, optimizing their individual fidelity. Such per-view supervision does not enforce geometric coherence among the remaining errors: where sparse observations provide weak constraints, predictions of the same physical point can drift differently across jointly generated views even when each appears plausible and incurs a small

per-view loss. A cross-view constraint is therefore needed, but directly matching RGB values, VAE latents, or velocities would be overly restrictive because viewpoint-dependent illumination, visibility, and local context legitimately alter these representations. We instead couple each prediction through its residual to its own target, enforcing consistency only in the error that should be removed.

For a sampled flow time, the predicted clean latent and its residual in query view j are

$$\hat{Z}_{0,j} = X_{\sigma,j} - \sigma V_{\theta,j}, \quad E_j = \hat{Z}_{0,j} - Z_{0,j}. \quad (6)$$

We reuse the VGGT- Ω point predictions to establish tracks across every pair of query views. A point is retained only when it projects inside both views, lies in front of both cameras, and passes a per-view z-buffer visibility test. Its projections are mapped to the nearest latent-grid locations p_a and p_b . For a valid track $c = (a, b, p_a, p_b)$, let $\Delta E_c = E_a(p_a) - E_b(p_b)$, and let $\bar{\rho}_\beta$ denote the channel average of the scalar Smooth-L1 penalty ρ_β . With normalized VGGT- Ω confidence w_c , PTRC is

$$\mathcal{L}_{\text{PTRC}} = \frac{\sum_{c \in \mathcal{C}} w_c \bar{\rho}_\beta(\Delta E_c)}{\sum_{c \in \mathcal{C}} w_c}, \quad \rho_\beta(x) = \begin{cases} x^2/(2\beta), & |x| < \beta, \\ |x| - \beta/2, & |x| \geq \beta. \end{cases} \quad (7)$$

The quadratic region provides precise alignment for small discrepancies, while the linear region, together with confidence weighting, limits the influence of unreliable tracks. Since PTRC aligns errors rather than predictions, corresponding views retain valid appearance changes without drifting independently from their respective targets.

3.4 TRAINING AND INFERENCE

All camera poses are expressed in a gauge defined solely by the source cameras. Consequently, changing the number or subset of query cameras does not alter the conditioning coordinate frame. We freeze the VAE and VGGT- Ω , and fine-tune the diffusion transformer together with the new input and routing modules. The number of jointly generated query views is varied during training to support both single-view synthesis and longer view trajectories.

Training Loss. The full objective combines query-wise flow matching with point-track regularization as $\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_{\text{PTRC}} \mathcal{L}_{\text{PTRC}}$, where $\lambda_{\text{PTRC}} = 0.1$ in all experiments. The two terms serve complementary roles: $\mathcal{L}_{\text{FM}}$ learns accurate denoising for each query view, while $\mathcal{L}_{\text{PTRC}}$ couples prediction errors along reliable 3D point tracks.

Geometry Condition Dropout. Geometry reconstructed from sparse observations is informative but inevitably imperfect. During training, we therefore randomly retain, attenuate, or remove the routed geometry condition, while leaving source appearance and camera rays unchanged. This prevents the diffusion model from treating projected geometry as an infallible scene reconstruction and improves its robustness when routed evidence is sparse, uncertain, or locally missing.

Geometry-Prior CFG. Geometry-condition dropout also provides a reference for selectively strengthening the geometry prior at inference. Let V_G denote the fully conditioned velocity prediction and V_{ref} the reference prediction obtained by removing only query-routed geometry. During early denoising, we apply $V_{\text{GeoCFG}} = V_{\text{ref}} + s_g(V_G - V_{\text{ref}})$, and otherwise retain V_G . The two branches share the diffusion state, source appearance, source-aligned visual features, camera conditions, timestep, and empty textual context. Thus, s_g strengthens query geometry without removing the observations that define the scene.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Data and training. We train on only 1K-scene split of DL3DV-10K (960P) (Ling et al., 2024). Removing one unavailable scene and 19 overlapping with the 140-scene DL3DV-Benchmark leaves 980 training scenes. VGGT-Diff initializes from Wan2.1-I2V-14B (Wan et al., 2025). We freeze the VAE, text and image encoders, and VGGT- Ω (Wang et al., 2026), while training the Wan DiT, expanded input projection, visual-feature adapter, and Visual Geometry Router. Training runs for 147 epochs at 192×336 , then 60 at 480×832 , with BF16, a global batch size of eight, and six source views. Each sample jointly denoises one to 16 targets. The Wan DiT uses learning rates of 10^{-5} and 5×10^{-6} in the two stages, respectively; new modules use 10^{-4} throughout.

Table 1: **Quantitative comparison of 6-view NVS on DL3DV and Mip-NeRF 360.** All methods follow the same protocol within each benchmark; #Scenes excludes generic foundation-model pre-training. LPIPS uses AlexNet features, and DS denotes DreamSim. Best, second, and third results are highlighted by **first**, **second**, and **third**.

Method	Backbone	#Scenes	DL3DV				Mip-NeRF 360			
			PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	DS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	DS $\downarrow$
<i>Regression-based Models</i>										
AnySplat (Jiang et al., 2025b)	VGGT (Wang et al., 2025)	254K	12.388	0.298	0.510	0.214	11.009	0.232	0.594	0.252
E-RayZer (Zhao et al., 2026)	–	10K	16.850	0.442	0.455	0.254	16.560	0.343	0.621	0.340
LVSM (Jin et al., 2024)	–	67.5K	17.090	0.478	0.333	0.204	15.250	0.317	0.609	0.577
DepthSplat (Xu et al., 2025)	Depth Anything V2 (Yang et al., 2024)	77.5K	17.284	0.566	0.307	0.158	15.977	0.370	0.410	0.215
<i>Diffusion-based Models</i>										
EscherNet (Kong et al., 2024)	SD1.5 (Rombach et al., 2022)	10K	12.070	0.251	0.484	0.227	11.140	0.126	0.540	0.315
Aether (Zhu et al., 2025)	CogVideoX-5B (Yang et al., 2025)	–	12.660	0.258	0.469	0.140	12.600	0.220	0.651	0.334
MVSplat360 (Chen et al., 2024b)	SVD (Blattmann et al., 2023)	69.5K	14.150	0.358	0.513	0.174	13.859	0.292	0.636	0.278
SEVA (Zhou et al., 2025)	SD2.1 (Rombach et al., 2022)	80K	16.150	0.470	0.253	0.088	14.590	0.294	0.372	0.137
FrameCrafter (Wu et al., 2026a)	Wan2.1-I2V-14B (Wan et al., 2025)	1K	17.180	0.445	0.223	0.066	15.640	0.279	0.365	0.111
VGGT-Diff	Wan2.1-I2V-14B (Wan et al., 2025)	1K	18.104	0.508	0.222	0.062	16.296	0.317	0.315	0.091

Table 2: **PSNR across target-pose difficulty on DL3DV.** All methods use independent 6-to-1 inference over 32 targets per bin.

Method	Interpolation			Extrapolation		
	Near	Mid	Far	Near	Mid	Far
<i>Regression-based Models</i>						
AnySplat (Jiang et al., 2025b)	14.196	12.415	10.862	13.977	12.566	11.827
E-RayZer (Zhao et al., 2026)	17.388	14.878	13.824	16.833	14.477	14.712
LVSM (Jin et al., 2024)	20.276	16.599	13.969	20.790	16.633	15.579
DepthSplat (Xu et al., 2025)	20.376	17.424	14.820	20.653	17.943	16.374
<i>Diffusion-based Models</i>						
Aether (Zhu et al., 2025)	14.030	12.167	11.065	15.739	12.295	11.795
GEN3C (Ren et al., 2025)	13.620	12.356	11.548	14.288	13.050	12.288
MVSplat360 (Chen et al., 2024b)	15.282	14.420	13.389	15.542	14.479	14.176
FrameCrafter (Wu et al., 2026a)	20.338	17.485	14.649	21.168	17.888	16.594
SEVA (Zhou et al., 2025)	20.940	17.861	14.703	22.311	18.050	16.575
<i>VGGT-Diff</i>	20.920	18.391	15.853	22.396	18.793	17.793

Table 3: **Ablations on DL3DV.** All variants use independent 6-to-1 inference on the 192 pose-stratified targets; LPIPS uses VGG features.

Configuration	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
<i>Training-time design</i>			
No VGGT- Ω prior	17.294	0.501	0.357
Point-rendered RGB	18.553	0.536	0.316
No PTRC	18.586	0.530	0.312
No condition regularization	18.722	0.538	0.309
Hard single-surface routing	18.895	0.548	0.303
Full (no CFG)	19.024	0.557	0.301
<i>Inference guidance</i>			
Full- Ω CFG ($s_g = 3$)	18.797	0.555	–
Target-only early CFG ($s_g = 2$)	19.139	0.563	0.296

Evaluation and baselines. We evaluate 6,188 DL3DV-Benchmark targets and test zero-shot transfer on Mip-NeRF 360 (Barron et al., 2022). All methods receive identical six-view sources and target cameras, with each target generated independently; *VGGT-Diff* uses 50 flow-sampling steps. Our evaluation follows FrameCrafter (Wu et al., 2026a), with native outputs resized or center-cropped to 480×480 . We report PSNR, SSIM (Wang et al., 2004), LPIPS-VGG (Zhang et al., 2018), and DreamSim (Fu et al., 2023). Baselines span regression-based and diffusion-based models in the tables. Additional diagnostics use 192 pose-stratified targets and eight-view geometric reconstruction.

4.2 COMPARISON WITH BASELINES

Overall performance. Table 1 compares all methods under the common protocol. On 6,188 DL3DV targets, *VGGT-Diff* ranks first in PSNR, LPIPS, and DreamSim and second in SSIM. Notably, these results are achieved using only $\sim 1k$ training scenes, substantially fewer than many competing methods, demonstrating strong data efficiency. Against FrameCrafter with the same Wan2.1-I2V-14B prior and a comparable training scale, *VGGT-Diff* gains 0.924 dB PSNR and 0.063 SSIM while retaining comparable perceptual quality, demonstrating the benefit of geometry-routed adaptation. On zero-shot Mip-NeRF 360, *VGGT-Diff* ranks first in LPIPS and DreamSim and second in PSNR. Although E-RayZer and DepthSplat lead PSNR and SSIM, respectively, their substantially higher perceptual errors indicate that these gains come at the cost of perceptual quality. **Qualitative comparison.** Figures 4 and 5 localize these gains. On DL3DV, *VGGT-Diff* preserves chair legs, desk boundaries, facade geometry, and legible signage while recovering target layout and occlusion order; competing methods blur thin structures, distort geometry, or drift from the requested camera. On zero-shot Mip-NeRF 360, *VGGT-Diff* likewise better preserves object boundaries and source-observed appearance, indicating scene-grounded rather than unsupported synthesis.

Pose robustness. Aggregate metrics can hide difficult-view failures. We divide 192 DL3DV targets into interpolation and extrapolation and further into near, mid, and far bins of 32 targets. Table 2 shows that *VGGT-Diff* ranks first in five bins and second in interpolation-near, only 0.020 dB behind SEVA. Its margin over FrameCrafter grows from 0.582 to 1.228 dB and exceeds 0.9 dB in five bins,

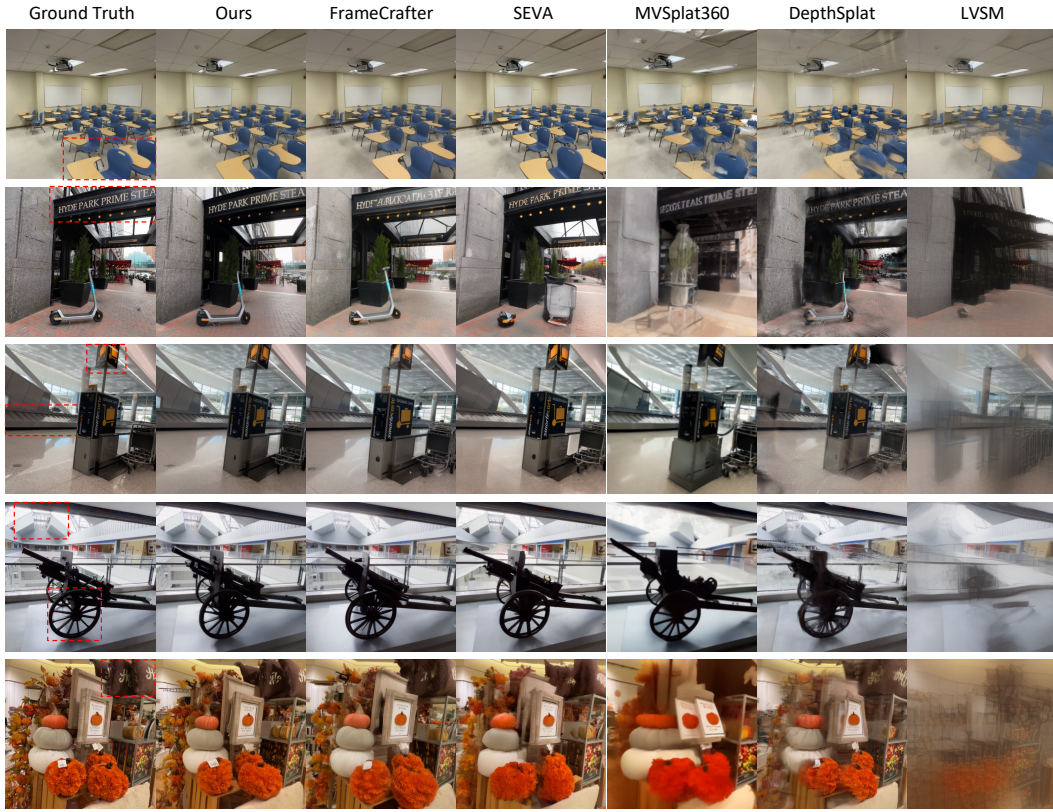


Figure 4: **Qualitative comparison on diverse DL3DV scenes.** *VGGT-Diff* better matches the requested viewpoints while preserving fine geometry, legible text, and object identity; previous methods exhibit blur, structural distortion, or content drift.

Table 4: **Point-cloud reconstruction quality on DL3DV.** Point clouds from eight jointly generated views are compared with GT-view references to assess cross-view consistency.

Method	Chamfer-L1 ↓	F-score@1% ↑	F-score@2% ↑
LVSM (Jin et al., 2024)	0.2387	0.1116	0.2218
SEVA (Zhou et al., 2025)	0.1944	0.2050	0.3244
FrameCrafter (Wu et al., 2026a)	0.1000	0.3086	0.4793
<i>VGGT-Diff</i>	0.0733	0.4231	0.6041

suggesting that explicit source-to-query conditioning becomes increasingly valuable outside dense source coverage. **Cross-view consistency.** Single-target metrics cannot determine whether jointly generated views support a coherent scene. Our diagnostic exploits the geometric prior of frozen *VGGT-Ω*: as a geometry foundation model, it consolidates mutually consistent multi-view evidence into coherent 3D structure. *Reconstructions that are more coherent and closer to those inferred from ground-truth views therefore indicate stronger cross-view consistency and rendering fidelity.*

Across 52 DL3DV scenes, all methods receive the same six sources and eight target cameras. We reconstruct their outputs and ground-truth views with *VGGT-Ω* under identical cameras, alignment, confidence filtering, and sampling. Chamfer-L1 and F-scores are normalized by the target-camera trajectory diameter to remove scale variation. Table 4 shows that *VGGT-Diff* reduces Chamfer-L1 by 26.7% relative to FrameCrafter and improves F-score@1% and F-score@2% by 11.45 and 12.48 percentage points. It wins the three metrics on 42, 44, and 43 of 52 scenes, respectively. Appendix Fig. 8 further shows fewer displaced points and more coherent structures. Since the references are reconstructed from target RGB rather than physical scans, these results measure relative consistency and reconstructability, not absolute 3D accuracy.

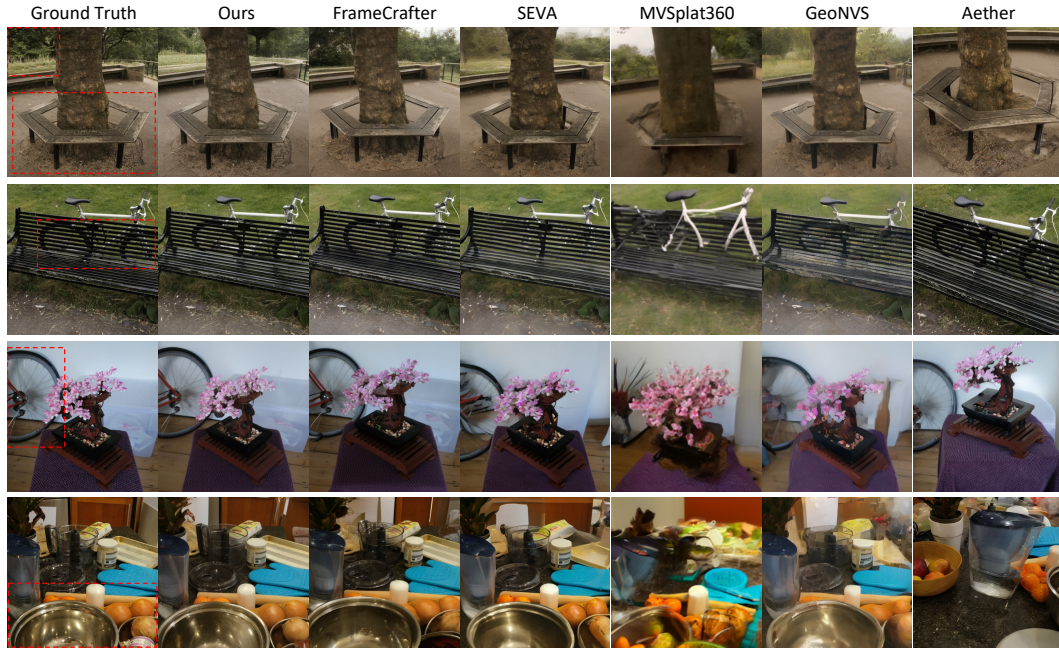


Figure 5: **Qualitative comparison on Mip-NeRF 360.** Without dataset-specific fine-tuning, *VGGT-Diff* better preserves fine structures, occlusion boundaries, and source-observed appearance while more accurately matching the target view.

4.3 ABLATION STUDIES

Protocol. All variants follow Sec. 4.1, start from the same Wan initialization, and change only the listed component. They are evaluated on the same 192 pose-balanced targets used above. Geometry-Prior CFG is disabled for training-time variants, isolating architectural and optimization choices. **Geometry conditioning.** Table 3 shows that removing all VGGT- Ω -dependent conditioning and regularization causes the largest degradation, reducing PSNR by 1.730 dB and SSIM by 0.056 while increasing LPIPS by 0.056. Camera rays and the Wan prior alone therefore cannot recover the target-view accuracy provided by visual geometry. Replacing routed visual features with point-rendered RGB, while retaining the same geometry, router, and PTRC, loses 0.471 dB PSNR and increases LPIPS by 0.015. The improvement thus comes from routing appearance-rich features rather than collapsing uncertain geometry into incomplete RGB renderings. **Consistency and routing.** Removing PTRC reduces PSNR by 0.438 dB and SSIM by 0.027, the largest SSIM drop among isolated component ablations. This confirms that per-view flow supervision alone leaves residual drift across corresponding scene points. Hard single-surface routing produces a smaller but consistent degradation, supporting the use of secondary-surface evidence under uncertain visibility. Removing geometry-condition regularization costs 0.302 dB PSNR and 0.019 SSIM while increasing LPIPS by 0.008, showing that stochastic attenuation and dropout reduce over-reliance on imperfect geometry. **Geometry-prior guidance.** The final rows reuse the Full checkpoint and alter only inference. Full- Ω CFG removes all VGGT- Ω conditions from its reference branch and guides throughout denoising, whereas target-only early CFG removes only query-routed geometry and acts during the high-noise stage. The latter improves the unguided model by 0.115 dB PSNR and 0.006 SSIM while reducing LPIPS by 0.005, with gains in five of six pose bins. Full- Ω CFG instead reduces PSNR by 0.227 dB, showing that the gain arises from matching the guidance scope and schedule to training-time geometry regularization rather than simply increasing guidance strength.

5 CONCLUSION

We presented *VGGT-Diff*, a geometry-routed multi-view framework combining visual geometry with pretrained video diffusion for sparse-view NVS. Its confidence-aware VGR maps source features into query-aligned conditions using 3D points and confidence, while PTRC aligns predicted-

clean residuals along reliable correspondences. Stochastic training-time weakening and selective inference-time strengthening improve robustness to imperfect geometry. Across diverse pose difficulties, *VGGT-Diff* achieves strong target-view fidelity and cross-view coherence, combining geometry-grounded scene structure with the generative prior’s ability to complete unseen content.

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A APPENDIX

Metric convention. The cross-method comparison in Table 1 uses LPIPS-AlexNet to follow the evaluation protocol of prior work. Controlled ablations use LPIPS-VGG consistently within the fixed 192-case diagnostic protocol. We label the backbone explicitly and do not compare absolute LPIPS values across these protocols.

A.1 VIEW SAMPLING AND POSE-DIFFICULTY PROTOCOL

Training view sampling. For each training scene, we precompute 16 reproducible six-view source bundles that rotate deterministically across epochs. Thirteen bundles use global sampling: the ordered camera trajectory is divided into six contiguous segments, and one frame is sampled from each segment. The remaining three use the same stratification within a randomly selected local window. The six selected views are shuffled before entering the model, exposing it to both broad trajectory coverage and locally concentrated observations.

Targets are sampled online after excluding the active source frames. Local bundles preferentially draw targets from their window and fall back to the full scene when necessary. The target-count curriculum progresses through $\{1, 2, 4\} \rightarrow \{2, 4, 8\} \rightarrow \{4, 8, 12\} \rightarrow \{4, 8, 12, 16\}$ in the first training stage and $\{4, 8\} \rightarrow \{4, 8, 12\} \rightarrow \{4, 8, 12, 16\}$ in the second. For eight or more targets, we always sample an ordered trajectory by selecting evenly spaced frames within a random temporal span. Smaller target sets use this mode with probability 0.5 and otherwise sample views uniformly without replacement. Training does not explicitly balance the pose-difficulty bins used for evaluation.

Source-defined camera gauge. Camera normalization depends only on the six source views. We choose the source camera nearest their centroid as the anchor, align all cameras to its rotation and origin, and normalize translation by the mean source-to-anchor distance. Query cameras do not affect this transformation, so single-target and joint inference share the same coordinate frame for a fixed source set.

Pose-difficulty definition. We construct the diagnostic set from the official six-view DL3DV-Benchmark split, using its fixed source views and candidate test frames. Let $i_{\min}$ and $i_{\max}$ be the minimum and maximum source frame indices. A target is trajectory interpolation when $i_{\min} < i_t < i_{\max}$ and trajectory extrapolation when it lies outside this interval. This definition follows the ordered capture trajectory and does not imply containment within the 3D convex hull of the source cameras.

Difficulty is measured by the target’s distance from its nearest source pose. Let $\mathbf{c}_t, \mathbf{f}_t$ denote the target camera center and unit forward direction, and $\mathbf{c}_j, \mathbf{f}_j$ those of source j . We define the source-trajectory diameter as $D_S = \max_{j,k \in \mathcal{S}} \|\mathbf{c}_j - \mathbf{c}_k\|_2$ and its stabilized value as $\bar{D}_S = \max(D_S, 10^{-6})$. The angular difference to source j is $\theta_j = \arccos[\text{clip}(\mathbf{f}_t^\top \mathbf{f}_j, -1, 1)]$. Pose novelty is then

$$\nu_t = \min_{j \in \mathcal{S}} \sqrt{\frac{\|\mathbf{c}_t - \mathbf{c}_j\|_2^2}{\bar{D}_S^2} + \frac{\theta_j^2}{\pi^2}}. \quad (8)$$

This combines scene-normalized translation and viewing-direction change. Near, mid, and far bins are defined by global pose-novelty tertiles computed separately for interpolation and extrapolation. Their two thresholds are (0.1165, 0.2147) for interpolation and (0.0560, 0.1272) for extrapolation.

Frozen evaluation cases. The candidate pool contains 5,549 interpolation and 638 extrapolation targets. From each of the six bins, we deterministically select 32 targets from 32 different scenes. The resulting protocol contains 192 unique targets from 109 scenes; scenes may recur across bins, but no scene-target pair is duplicated. Every method receives the same six source views and predicts each target independently using fixed per-case noise, without additional target slots, rollout, or prediction feedback. This pose-stratified diagnostic is separate from the aggregate FrameCrafter-aligned comparison in Table 1, and its cases remain fixed across model resolutions.

A.2 SCALING WITH TRAINING DATA AND OPTIMIZATION BUDGET

We evaluate data and optimization scaling under the same half-resolution 192-case protocol used for controlled ablations. At 18K updates, increasing the clean training set from 980 scenes to 10K

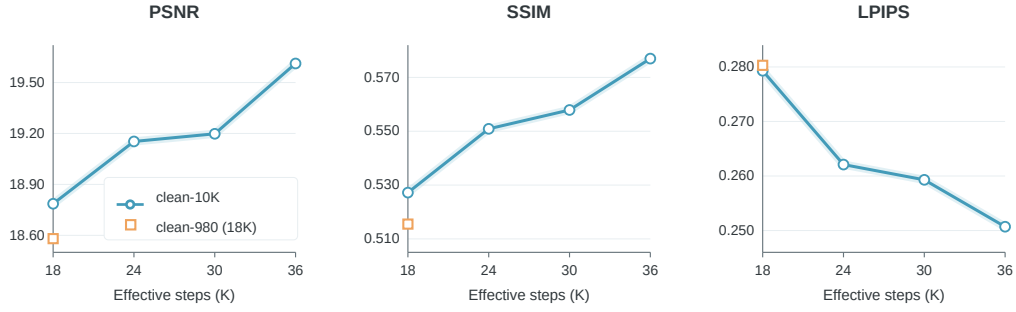


Figure 6: **Scaling with training data and optimization budget.** The clean-10K model is evaluated from 18K to 36K effective updates; the isolated clean-980 marker provides the matched 18K reference. LPIPS uses VGG features.

Table 5: **Joint versus independent eight-view generation.** Image metrics use 416 targets; geometry proxies are computed from the same outputs with frozen VGGT- Ω . LPIPS uses VGG features, and Chamfer-L1 is trajectory-normalized.

Mode	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Chamfer $\downarrow$	F@1% $\uparrow$	F@2% $\uparrow$
Joint-8	17.6401	0.4787	0.3568	0.0483	0.3438	0.5341
Independent-8	17.1784	0.4453	0.3744	0.0741	0.2432	0.4047

improves PSNR from 18.581 to 18.786 dB and SSIM from 0.5155 to 0.5272, while reducing LPIPS-VGG from 0.2803 to 0.2793. This matched-budget comparison isolates the effect of additional training scenes.

With additional optimization, the clean-10K model reaches 19.612 dB PSNR, 0.5770 SSIM, and 0.2507 LPIPS-VGG at 36K updates. The monotonic overall trend indicates that the larger training set remains under-optimized at 18K. However, the 24K–36K improvements combine data scale with additional updates; they should not be attributed to either factor alone until the corresponding clean-980 compute-matched checkpoints are available.

A.3 BENEFITS OF JOINT MULTI-TARGET GENERATION

We compare generating eight targets in one diffusion sequence (Joint-8) with eight strictly independent 6-to-1 runs. Across 52 DL3DV scenes and 416 targets, both modes use the same checkpoint, six sources, eight target cameras, seed, per-view noise, and 50-step sampler. Independent runs cannot access other target cameras, latents, or predictions.

Joint generation improves PSNR by 0.462 dB and SSIM by 0.0334, reduces LPIPS-VGG by 0.0176, lowers normalized Chamfer-L1 by 0.0259, and raises F-score@1% and F-score@2% by 0.1006 and 0.1294. Scene-bootstrap 95% confidence intervals exclude zero for all three image and geometry metrics. Joint target slots therefore provide useful mutual context rather than merely batching otherwise independent predictions.

A.4 VGGT- Ω DESIGN CHOICES FOR GEOMETRY CONDITIONING

Setup. We examine feature depth and optimization scope under the same half-resolution 192-case protocol. First, we vary the extraction layer while keeping VGGT- Ω frozen. Second, using Layer 23, we jointly optimize its visual aggregator and encoder at a learning rate of 10^{-6} while retaining frozen camera and depth heads. The Wan DiT, adapter, and Visual Geometry Router are trained normally in all variants; all remaining inputs and objectives are fixed.

Feature depth. Table 6 shows a modest preference for intermediate features. Layer 11 performs best across all metrics, improving over Layer 23 by 0.153 dB PSNR and 0.0098 SSIM while reducing LPIPS-VGG by 0.0040. Intermediate representations may better balance cross-view correspondence with local appearance: shallower features contain less aggregated multi-view context,

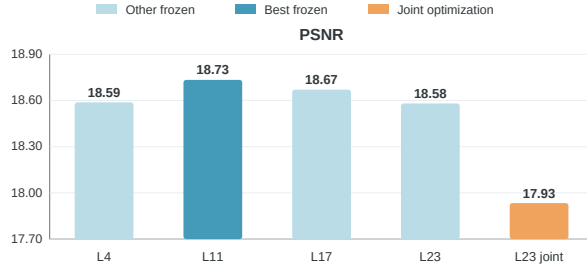


Figure 7: **VGGT- Ω feature depth and optimization.** Intermediate frozen features perform best, while jointly optimizing the Layer 23 visual pathway degrades PSNR.

Table 6: **VGGT- Ω feature and optimization diagnostics.** All variants use the same half-resolution protocol; LPIPS uses VGG features.

Configuration	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
<i>Frozen VGGT-Ω: feature depth</i>			
Layer 4	18.588	0.5160	0.2828
Layer 11	18.734	0.5253	0.2763
Layer 17	18.670	0.5197	0.2788
Layer 23	18.581	0.5155	0.2803
<i>Additional optimization diagnostic</i>			
Layer 23, jointly optimized	17.934	0.4803	0.3021

whereas deeper features may trade fine detail for more abstract scene information. The limited variation among frozen layers also indicates robustness to feature depth.

Frozen versus jointly optimized features. Joint optimization reduces PSNR by 0.647 dB and SSIM by 0.0352, while increasing LPIPS-VGG by 0.0218 relative to the frozen Layer 23 setting. Under the current clean-980 data, learning rate, and frozen-head configuration, additional trainable capacity therefore does not improve NVS. One plausible explanation is that flow-matching gradients alter the pretrained visual representation while the frozen geometry heads retain their original feature organization, producing internal drift.

These half-resolution results are not directly comparable with the full-resolution main results. All main-paper experiments retain the pre-specified frozen Layer 23 configuration to avoid retrospective model selection on the fixed evaluation set. The complementary behavior across depths motivates multi-level feature fusion as a structured alternative that preserves the pretrained geometry representation while combining local appearance with broader multi-view context.

A.5 COMPLETE POSE-DIFFICULTY RESULTS

We provide complete PSNR and LPIPS results across interpolation and extrapolation difficulty. All methods use six fixed source views and independent 6-to-1 inference on a common 480×480 image plane, with bins determined solely by camera pose. Each DL3DV bin contains 32 targets; Mip-NeRF 360 contains 66 targets per interpolation bin and 16 per extrapolation bin. Ranking highlights follow Table 1, and dashes denote unavailable results.

Unlike the aggregate comparison in Table 1, this diagnostic separates trajectory relation from pose novelty. Interpolation and extrapolation are each divided into near, mid, and far bins, revealing whether performance degrades smoothly as target cameras move away from the available observations. The same frozen cases are used for every method, so differences across bins reflect pose robustness rather than sampling changes.

Analysis. On DL3DV, *VGGT-Diff* achieves the best PSNR in five of six bins and the best LPIPS in all six, while trailing SEVA by only 0.020 dB in interpolation-near. Its PSNR margin over Frame-Crafter increases from 0.582 dB in interpolation-near to 1.228 dB in extrapolation-far, indicating that explicit source-to-query geometry becomes more valuable as the requested pose departs from source coverage. On zero-shot Mip-NeRF 360, *VGGT-Diff* obtains the best LPIPS in every bin and

Table 7: **PSNR across target-pose difficulty ($\uparrow$)**. Results are reported separately for DL3DV and Mip-NeRF 360.

Method	DL3DV						Mip-NeRF 360					
	Interpolation			Extrapolation			Interpolation			Extrapolation		
	Near	Mid	Far	Near	Mid	Far	Near	Mid	Far	Near	Mid	Far
<i>Regression-based Models</i>												
AnySplat (Jiang et al., 2025b)	14.196	12.415	10.862	13.977	12.566	11.827	12.424	11.040	9.851	12.594	10.273	8.972
E-RayZer (Zhao et al., 2026)	17.388	14.878	13.824	16.833	14.477	14.712	15.964	15.085	14.028	15.404	14.760	12.609
LVSM (Jin et al., 2024)	20.276	16.599	13.969	20.790	16.633	15.579	16.117	14.488	13.722	15.296	14.040	13.016
DepthSplat (Xu et al., 2025)	20.376	17.424	14.820	20.653	17.943	16.374	17.478	16.046	15.303	16.208	15.518	12.511
<i>Diffusion-based Models</i>												
Aether (Zhu et al., 2025)	14.030	12.167	11.065	15.739	12.295	11.795	12.977	12.994	12.867	11.841	10.873	10.894
GEN3C (Ren et al., 2025)	13.620	12.356	11.548	14.288	13.050	12.288	—	—	—	—	—	—
MVSplat360 (Chen et al., 2024b)	15.282	14.420	13.389	15.542	14.479	14.176	14.283	13.928	13.585	14.407	13.463	12.794
FrameCrafter (Wu et al., 2026a)	20.338	17.485	14.649	21.168	17.888	16.594	16.933	15.700	15.016	16.222	15.330	13.017
SEVA (Zhou et al., 2025)	20.940	17.861	14.703	22.311	18.050	16.575	17.911	16.467	15.427	17.448	15.925	13.933
VGGT-Diff	20.920	18.391	15.853	22.396	18.793	17.793	17.254	16.630	15.611	16.336	16.061	13.989

Table 8: **LPIPS across target-pose difficulty ($\downarrow$)**. We use VGG features on DL3DV and AlexNet features on Mip-NeRF 360.

Method	DL3DV						Mip-NeRF 360					
	Interpolation			Extrapolation			Interpolation			Extrapolation		
	Near	Mid	Far	Near	Mid	Far	Near	Mid	Far	Near	Mid	Far
<i>Regression-based Models</i>												
AnySplat (Jiang et al., 2025b)	0.5052	0.5609	0.6214	0.5057	0.5483	0.5828	0.5513	0.5854	0.6135	0.5920	0.6349	0.6830
E-RayZer (Zhao et al., 2026)	0.5496	0.6357	0.6925	0.5527	0.6433	0.6508	0.7270	0.7978	0.8284	0.7494	0.7975	0.7859
LVSM (Jin et al., 2024)	0.2911	0.4413	0.5656	0.2375	0.3913	0.4835	0.4980	0.6461	0.6897	0.5315	0.6589	0.7058
DepthSplat (Xu et al., 2025)	0.2688	0.3592	0.4643	0.2600	0.3141	0.4077	0.3164	0.4058	0.4555	0.3965	0.4842	0.5703
<i>Diffusion-based Models</i>												
Aether (Zhu et al., 2025)	0.4949	0.5891	0.6405	0.3952	0.5579	0.6098	0.5731	0.5855	0.5978	0.5965	0.6857	0.6980
GEN3C (Ren et al., 2025)	0.5194	0.5808	0.6269	0.4773	0.5578	0.5990	—	—	—	—	—	—
MVSplat360 (Chen et al., 2024b)	0.5009	0.5374	0.5952	0.4855	0.5223	0.5724	0.5820	0.6453	0.6610	0.6404	0.6683	0.6743
FrameCrafter (Wu et al., 2026a)	0.2489	0.3558	0.4689	0.2150	0.3011	0.3996	0.2712	0.3639	0.3996	0.3148	0.4204	0.5152
SEVA (Zhou et al., 2025)	0.2460	0.3390	0.4546	0.2122	0.2959	0.3929	0.2564	0.3267	0.3854	0.2983	0.4013	0.4744
VGGT-Diff	0.2423	0.3227	0.4163	0.1922	0.2732	0.3572	0.2474	0.3085	0.3504	0.2922	0.3685	0.4424

the best PSNR in four of six bins. The consistent perceptual gains across both datasets suggest that the geometry-routed diffusion prior remains scene-grounded under increasing viewpoint change.

A.6 POINT-CLOUD RECONSTRUCTION

Figure 8 complements the quantitative geometry evaluation with point clouds reconstructed from jointly generated views. Point-to-reference errors expose geometric disagreement that may be less apparent in individual rendered images.

This diagnostic treats frozen VGGT- Ω as a common geometry probe. A coherent set of views provides mutually compatible evidence and can therefore be consolidated into a stable point cloud, whereas cross-view drift produces duplicated, displaced, or fragmented structure. All methods use the same six source views and eight target cameras, and their reconstructions follow the same alignment, confidence filtering, and sampling procedure. The GT-view reconstruction provides a common reference rather than an absolute 3D scan.

The error maps in Fig. 8 localize the differences summarized by Table 4. SEVA and FrameCrafter exhibit larger high-error regions and displaced foreground points, while VGGT-Diff produces a more compact reconstruction with fewer structural offsets. The corresponding rendered point cloud also better preserves the shared layout across views, supporting the quantitative gains in Chamfer-L1 and F-score.

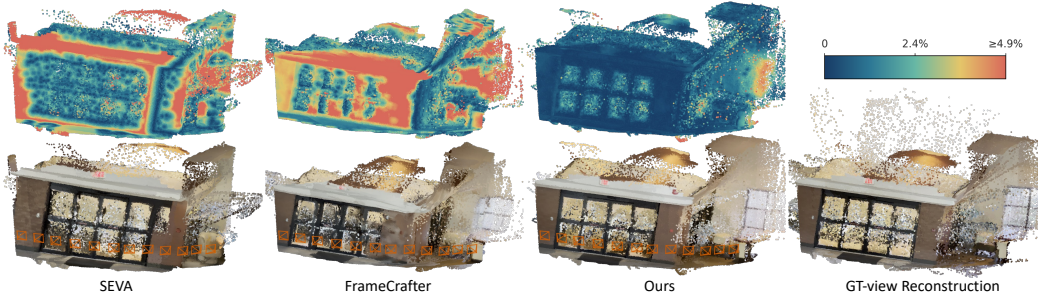


Figure 8: **Point-cloud reconstruction from jointly generated views.** Normalized point-to-reference errors are shown above the corresponding reconstructions. *VGGT-Diff* produces fewer displaced points and more coherent geometry, indicating stronger cross-view consistency.

Table 9: **PTRC versus direct velocity matching across pose difficulty.** Both use the same half-resolution protocol; PTRC values are highlighted and LPIPS uses VGG features.

Interpolation						Extrapolation					
Difficulty	PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$		Difficulty	PSNR $\uparrow$		SSIM $\uparrow$	
	Direct	PTRC	Direct	PTRC	Direct	PTRC		Direct	PTRC	Direct	PTRC
Near	19.008	20.133	0.531	0.602	0.274	0.221	Near	20.021	21.326	0.591	0.674
Mid	17.520	18.103	0.435	0.467	0.341	0.300	Mid	17.863	18.613	0.478	0.535
Far	15.210	15.670	0.308	0.352	0.440	0.397	Far	16.994	17.641	0.415	0.463

A.7 RESIDUAL VERSUS RAW-VELOCITY CONSISTENCY

Motivation. PTRC regularizes predicted-clean residuals rather than directly tying raw model outputs across views. To isolate this design choice, we replace PTRC with a naive objective that applies the same point tracks, confidence weights, and robust penalty directly to predicted velocities:

$$\mathcal{L}_{\text{Vel}} = \frac{\sum_{c \in \mathcal{C}} w_c \bar{\rho}_{\beta}(V_{\theta,a}(p_a) - V_{\theta,b}(p_b))}{\sum_{c \in \mathcal{C}} w_c}. \quad (9)$$

This alternative assumes that corresponding locations should share the same velocity. Under linear flow matching, however, each view has its own target $U_j = \epsilon_j - Z_{0,j}$; view-specific clean latents and noise states therefore make raw velocity equality generally invalid. Since $E_j = -\sigma(V_{\theta,j} - U_j)$, PTRC instead aligns errors relative to each view’s own flow target, preserving valid differences between their velocities.

This comparison is stricter than simply removing PTRC. The direct-matching baseline retains the same 3D correspondences and explicitly couples the same query locations, but changes what is required to agree. It therefore tests whether the benefit comes from generic point-track smoothing or specifically from aligning residual errors relative to each view’s supervision target.

Controlled comparison. Both variants share the data, initialization, geometry conditioning, point-track construction, and optimization, differing only in the consistency objective. For efficiency, they follow the same half-resolution protocol and are evaluated on the same 192 pose-stratified targets. Geometry conditioning, condition regularization, target curriculum, and robust penalty remain unchanged; both consistency objectives use weight $\lambda_{\text{PTRC}} = 0.1$.

Analysis. Across all 192 targets, PTRC improves PSNR from 17.769 to 18.581 dB and SSIM from 0.460 to 0.516, while reducing LPIPS from 0.330 to 0.280. Table 9 further shows consistent gains on every metric across all six pose bins, with PSNR improvements ranging from 0.460 to 1.305 dB. Because both variants use identical correspondences and weighting, these gains isolate the residual-relative formulation rather than the point tracks themselves. Direct velocity matching penalizes valid differences between view-specific flow targets, whereas PTRC couples only the prediction errors that should be removed. The improvement appears in both interpolation and extrapolation, showing that the residual formulation is not tied to one camera relation or difficulty range.

Table 10: **Effect of Wan adaptation strategy.** Both variants use the same half-resolution controlled protocol; LPIPS uses VGG features.

Wan adaptation	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
LoRA, rank 32	17.2563	0.4677	0.3554
Full fine-tuning	18.5807	0.5155	0.2803

A.8 ADDITIONAL OPTIMIZATION AND STATISTICAL DIAGNOSTICS

Wan adaptation. We compare full DiT fine-tuning with a rank-32 LoRA adaptation while retaining the same geometry conditioner, data, initialization, objectives, and half-resolution 192-case evaluation. Full fine-tuning improves PSNR by 1.324 dB and SSIM by 0.0478, while reducing LPIPS-VGG by 0.0751. The large gap indicates that sparse low-rank updates are insufficient to integrate routed geometry into the pretrained video prior under this setup.

Paired scene-level statistics. We additionally compare locally generated *VGGT-Diff* and FrameCrafter outputs under the same 140-scene, 6,188-target protocol. The scene-mean PSNR gain is 0.7893 dB, with a 92.14% scene win rate and a scene-bootstrap 95% confidence interval of [0.7003, 0.8769] dB. The corresponding scene-mean changes are +0.0330 SSIM, −0.0365 LPIPS-AlexNet, and −0.0118 DreamSim. These paired results show that the aggregate improvement is distributed across scenes rather than driven by a small subset. They are reported separately from Table 1, whose baseline rows follow the published cross-method protocol.